Heaps

Joseph Chuang-Chieh Lin (林莊傑)

Department of Computer Science & Engineering,
National Taiwan Ocean University

Fall 2026



Outline

1. Introduction
   - Building a heap




Outline

1. Introduction
   - Building a heap




Heaps

Max Tree

A max tree is a tree in which

- the key value in each node $\geq$ the key values in its children.



Heaps

Max Tree

A max tree is a tree in which

- the key value in each node $\geq$ the key values in its children.

Min Tree

A min tree is a tree in which

- the key value in each node $\leq$ the key values in its children.




Heaps

Max Tree

A max tree is a tree in which

- the key value in each node $\geq$ the key values in its children.

Min Tree

A min tree is a tree in which

- the key value in each node $\leq$ the key values in its children.

Max Heap

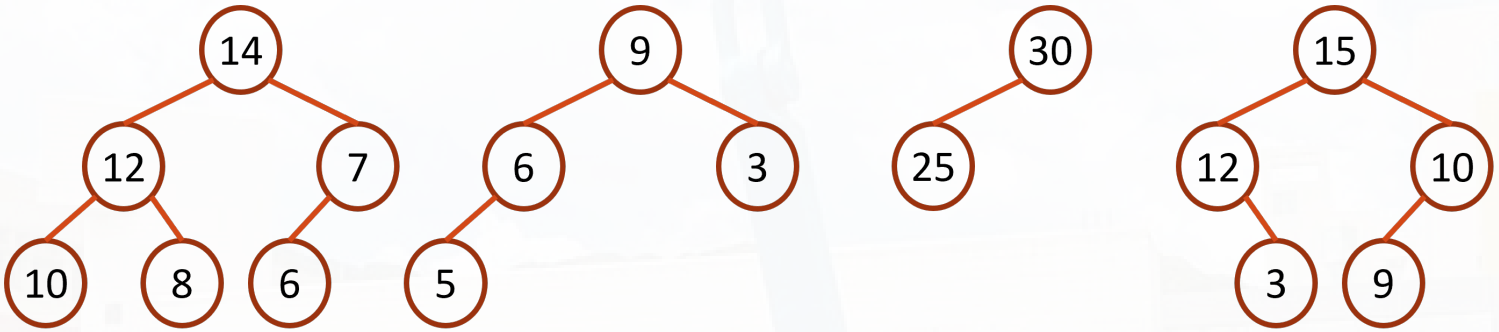
A complete binary tree that is also a max tree.

Min Heap

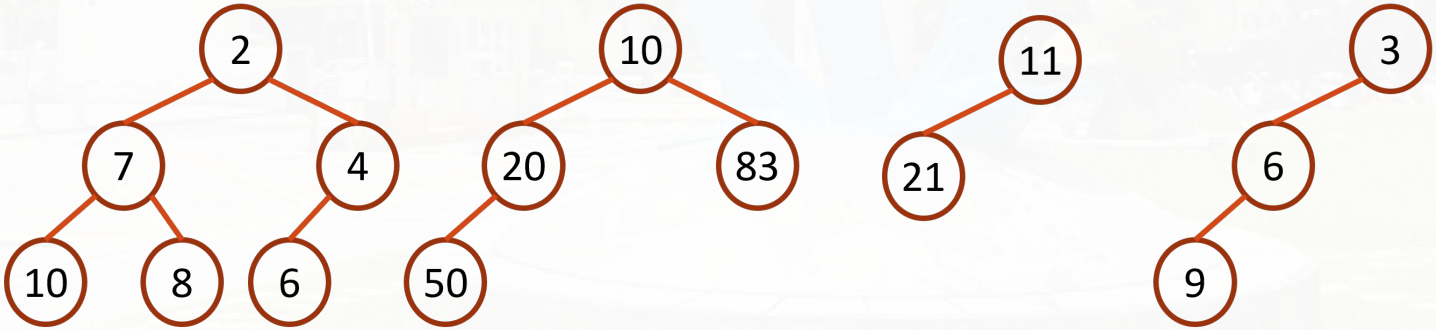
A complete binary tree that is also a min tree.




Examples: Max & Min Trees

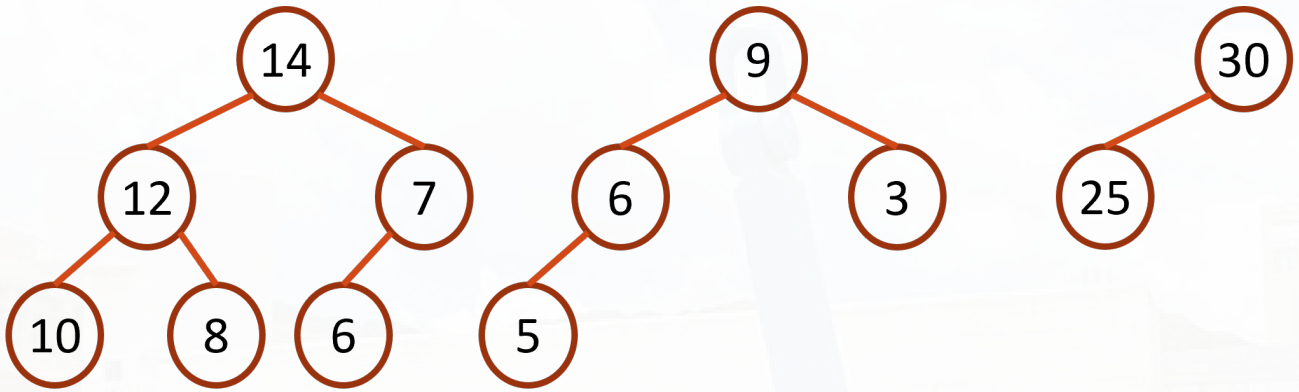


Max Trees

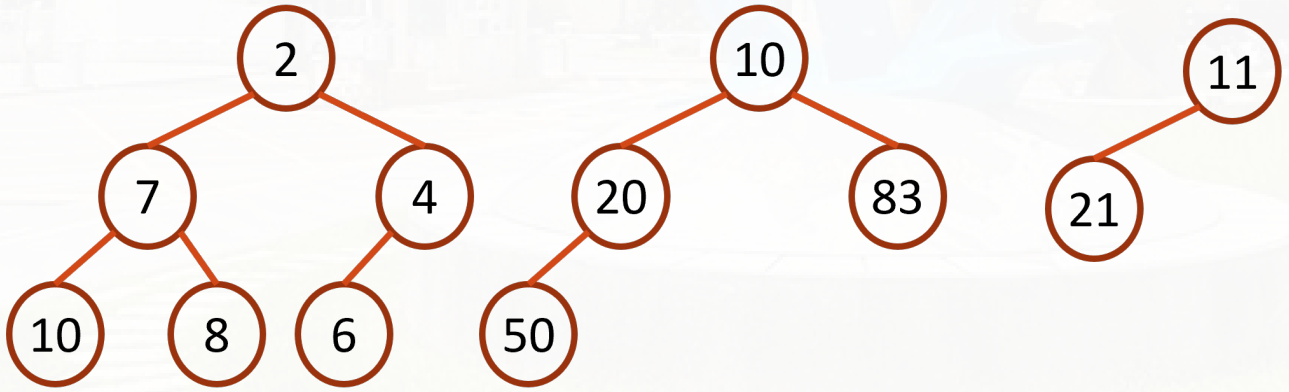


Min Trees

Examples: Max & Min Heaps



Max Heaps



Min Heaps

The Key Application: Priority Queues

- Heaps are frequently used to implement priority queues.
- In this kind of queue,
  - the element to be deleted is the one with highest (or lowest) priority.
  - at **any time**, an element with arbitrary priority can be **inserted** into the queue.



Some Important Notes

- It's straightforward to implement a heap using an array (WHY?).



Some Important Notes

- It's straightforward to implement a heap using an array (WHY?).
- Insert the new node next to the last element in the array.



Some Important Notes

- It's straightforward to implement a heap using an array (WHY?).
- Insert the new node next to the last element in the array.
- A heap is a complete binary tree.



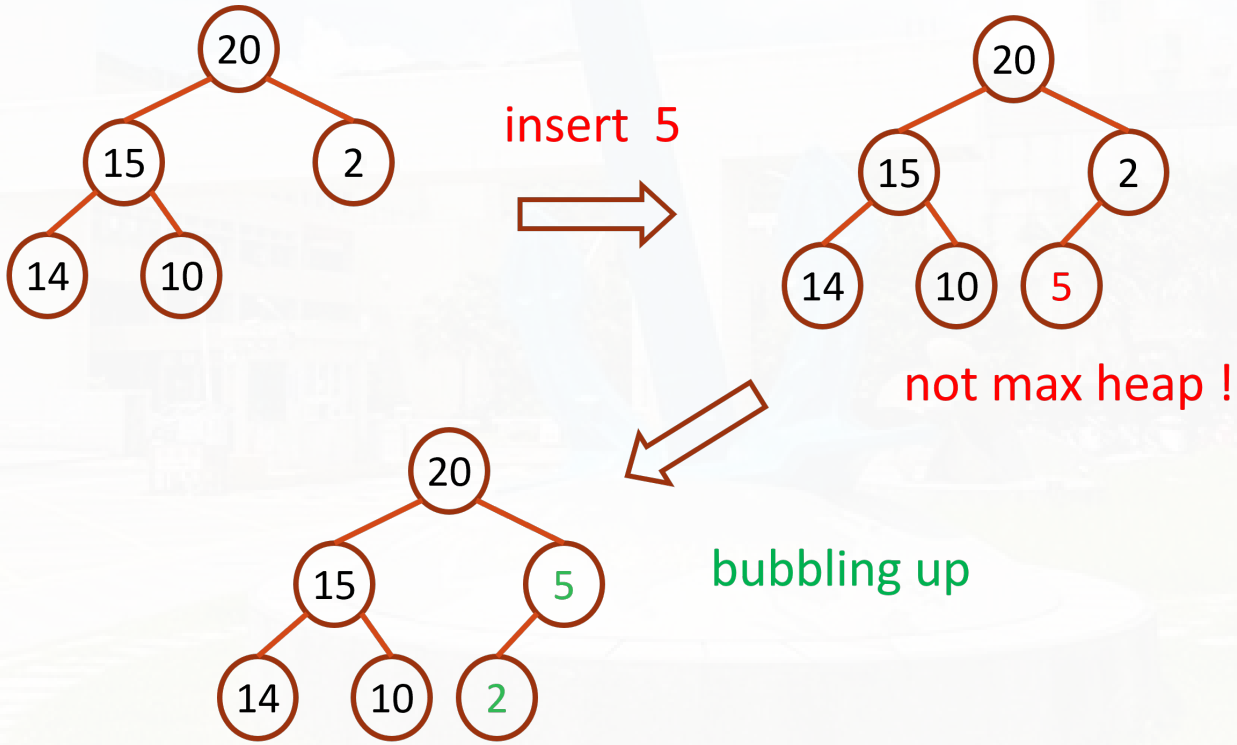
Insertion into a Max Heap

- The bubbling process.
  - It begins at the new node of the tree and moves toward the root.



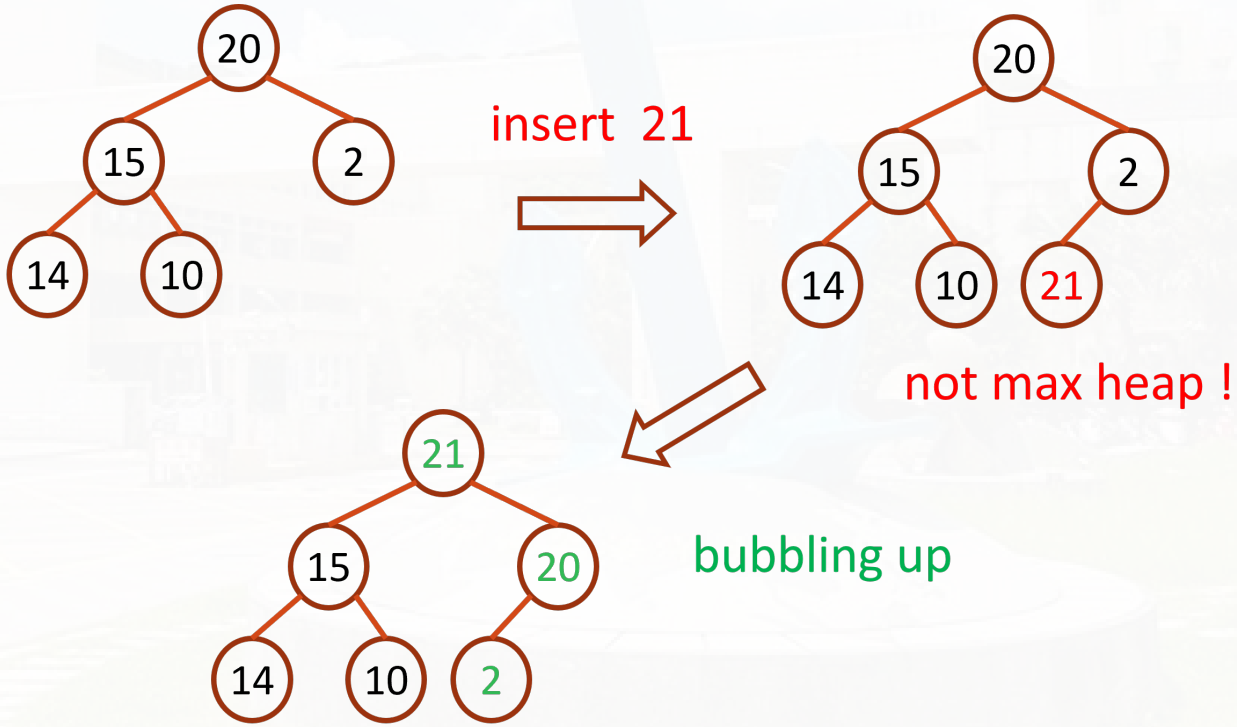
Insertion into a Max Heap

- The bubbling process.
  - It begins at the new node of the tree and moves toward the root.



Insertion into a Max Heap

- The bubbling process.
  - It begins at the new node of the tree and moves toward the root.



The Code for Insertion into a Max Heap

- Consider the following declarations:

```
#define MAX_ELEMENTS 200  /* maximum heap size+1 */
#define HEAP_FULL (n) (n == MAX_ELEMENTS -1)
#define HEAP_EMPTY (n)  (!n)
typedef struct {
    int key;
    /* other fields */
} element;
element heap[MAX_ELEMENTS];
int n = 0;
```



The Code for Insertion into a Max Heap

```
void push (element item, int *n) {
/* insert item into a max heap of current size *n */
    int i;
    if (HEAP_FULL(*n)) {
        printf("The heap is full.\n");
        exit(EXIT_FAILURE);
    } // O(1) time
    i = ++(*n); // the insertion index is n+1
    /* try bubbling up if necessary */
    while ((i != 1) && (item.key > heap[i/2].key)) {
        heap[i] = heap[i/2];
        i /= 2;
    } //  O(lg n) time
    heap[i] = item; // O(1) time
}
```

- Calling `push(item, &n)`
- The time complexity of the insertion: $O(\lg n)$.




Deletion from a Max Heap

function call: `pop(&n)`

- When an element is to be deleted from a max heap, it is ALWAYS taken from the root of the heap.



Deletion from a Max Heap

function call: `pop(&n)`

- When an element is to be deleted from a max heap, it is ALWAYS taken from the root of the heap.
- The steps of deletion from a Max heap:
  - delete the root node.
  - insert the last node into the root (say r).
  - use the bubbling up process to ensure that the resulting heap remains a max heap (a.k.a. heapify at r).



Illustration of Deletion from a Max Heap

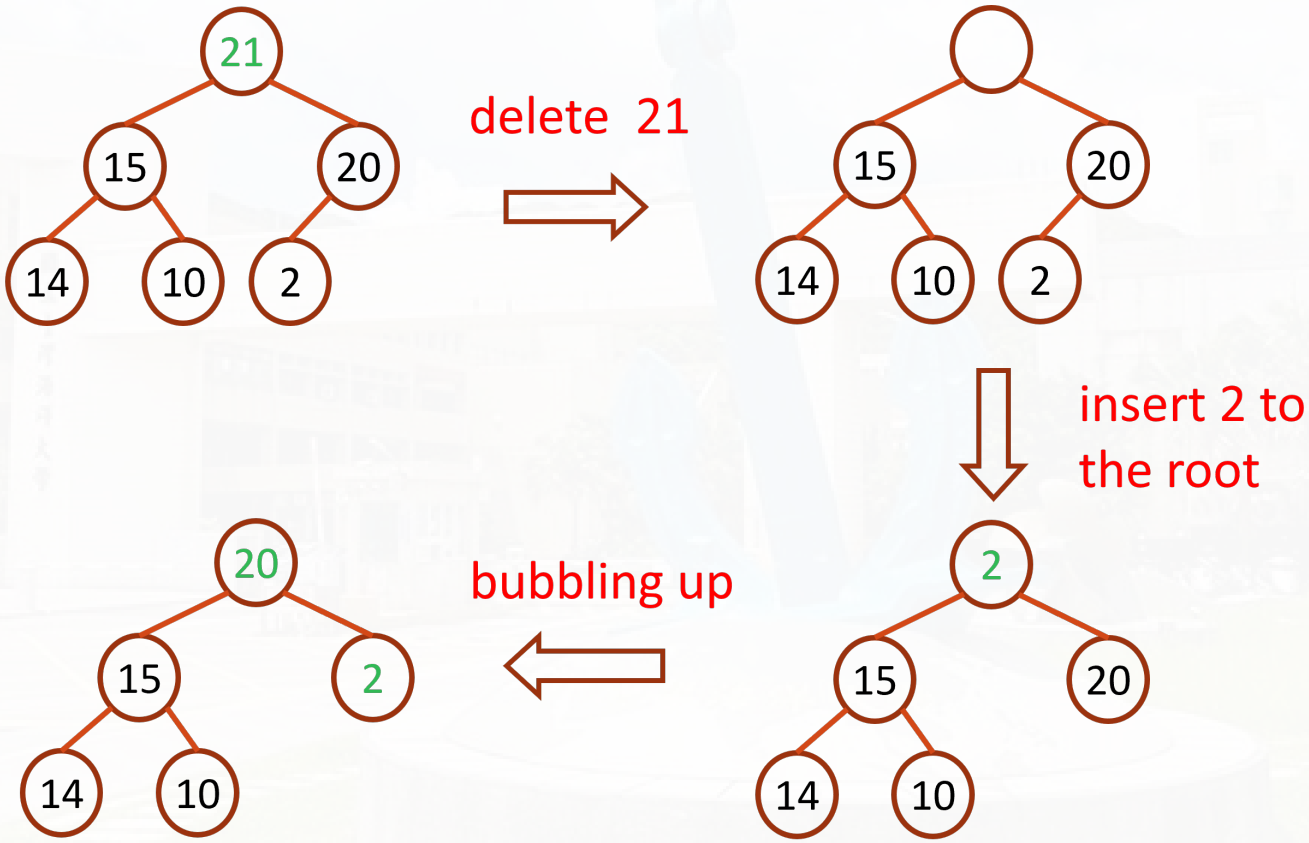
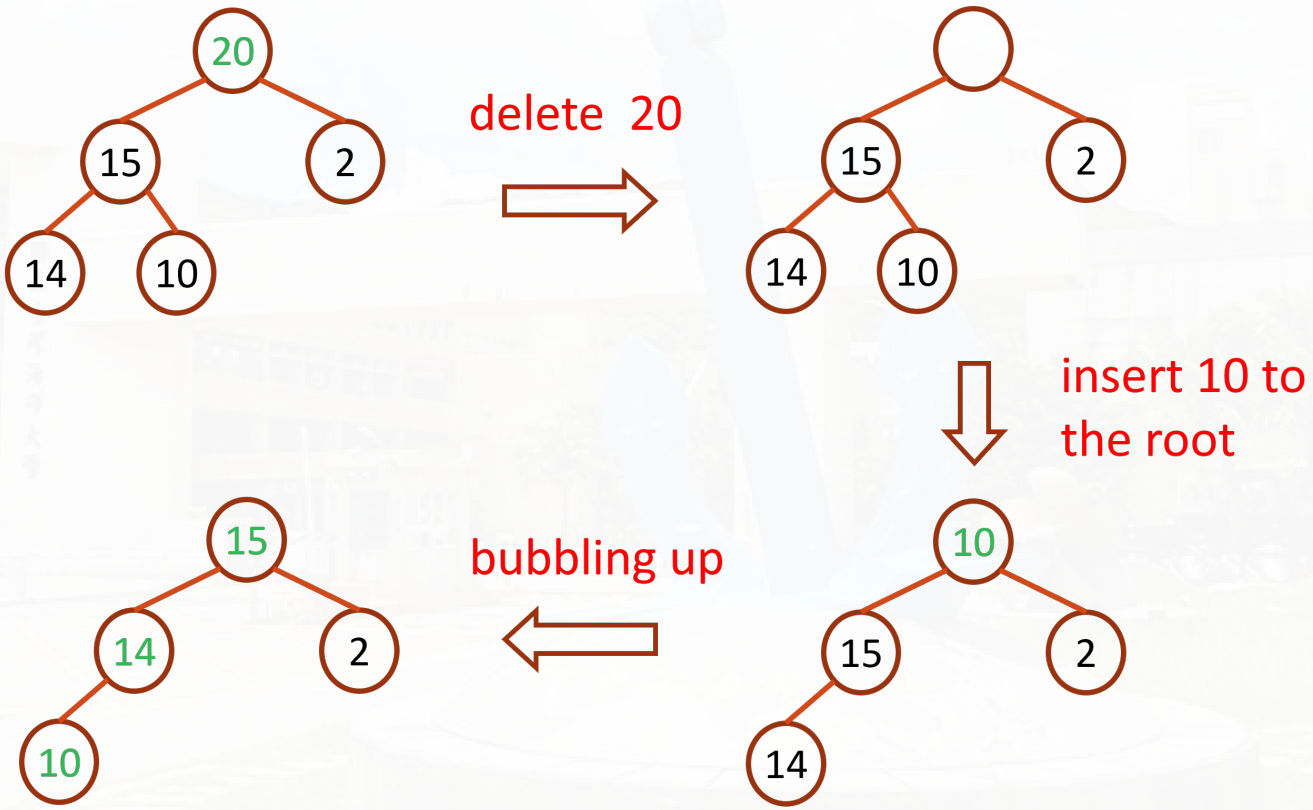


Illustration of Deletion from a Max Heap



The Code for Deletion from a Max Heap

```
element pop(int *n) {
/* delete element with the highest key from the heap */
    int parent, child;
    element item, temp;
    if (HEAP_EMPTY(*n)) {
        fprintf(stderr, "The heap is empty\n");
        exit(EXIT_FAILURE);
    }
    /* save value of the element with the highest key */
    item = heap[1];
    /* use last element in heap to adjust heap */
    temp = heap[(*n)--]; // start Heapify from the root
    parent = 1;
    child = 2; // default: the left child
    while (child <= *n) { // O(lg n) time
    /* find the larger child of the current parent */
        if ((child < *n) && (heap[child].key < heap[child+1].key))
            child++; // okay, it's the right child!
        if (temp.key >= heap[child].key) break; // the new root is the maximum!
        /* if the max-child gets larger key, move to the next lower level */
        heap[parent] = heap[child];
        parent = child;
        child *= 2;
    }
    heap[parent] = temp;
    return item;
}
```




Time Complexity of the Deletion from a Max Heap

- Delete the root node: $O(1)$.
- Insert the last node to the root: $O(1)$.
- Since the height of the heap is $\lceil \lg(n+1) \rceil$, the while loop is iterated for $O(\lg n)$ times.
- Thus, the overall time complexity: the time complexity of the deletion: $O(\log n)$.



Outline

1. Introduction
   - Building a heap




How to build a heap for a set of n input numbers?

- For each input number x, execute `push`(x).




How to build a heap for a set of n input numbers?

- For each input number x, execute `push(x)`.

- The above process is correct and requires



How to build a heap for a set of n input numbers?

- For each input number x, execute `push(x)`.

- The above process is correct and requires $O(n \log n)$ time.



An $O(n)$ time algorithm for building a (max) heap

Input: n numbers: $x_1, x_2, \ldots, x_n$.

Efficient Heap Construction

1. For each input number x_i, insert x_i into array A at $A[i-1]$ one by one.
2. For $i = \lfloor n/2 \rfloor - 1$ down to 0:
   - Run `heapify`(A, n, i)



An $O(n)$ time algorithm for building a (max) heap

Input: n numbers: $x_1, x_2, \ldots, x_n$.

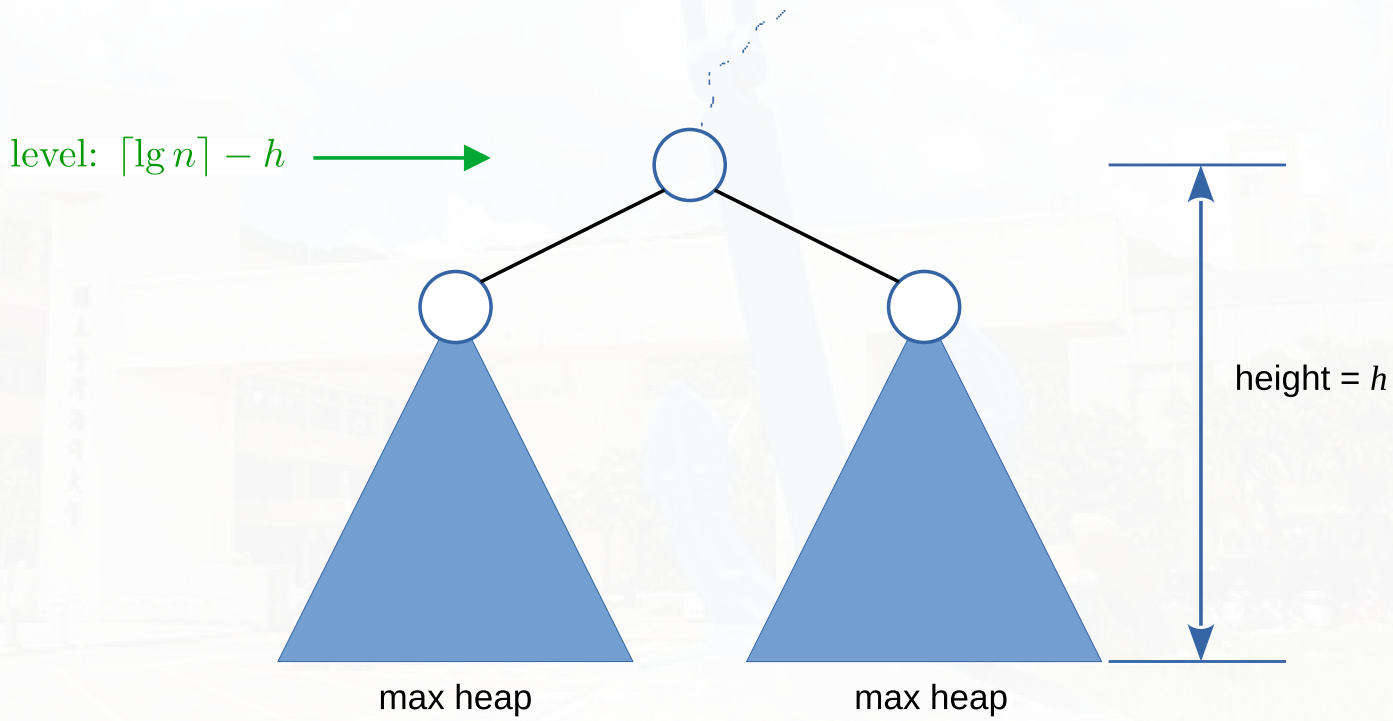
Efficient Heap Construction

1. For each input number x_i, insert x_i into array A at $A[i-1]$ one by one.
2. For $i = \lfloor n/2 \rfloor - 1$ down to 0:
   - Run `heapify`(A, n, i)

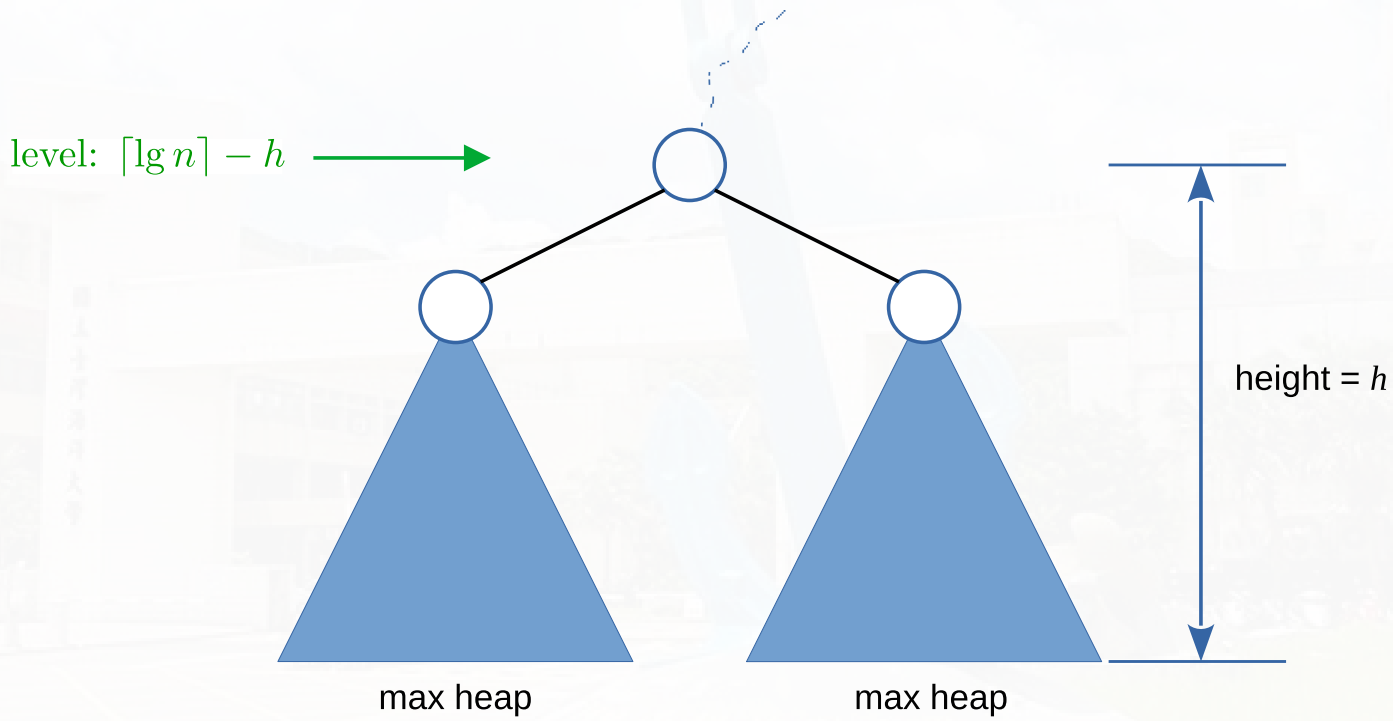
- That is, we build a heap in a bottom-up fashion!




Heap recursive view (bottom-up)

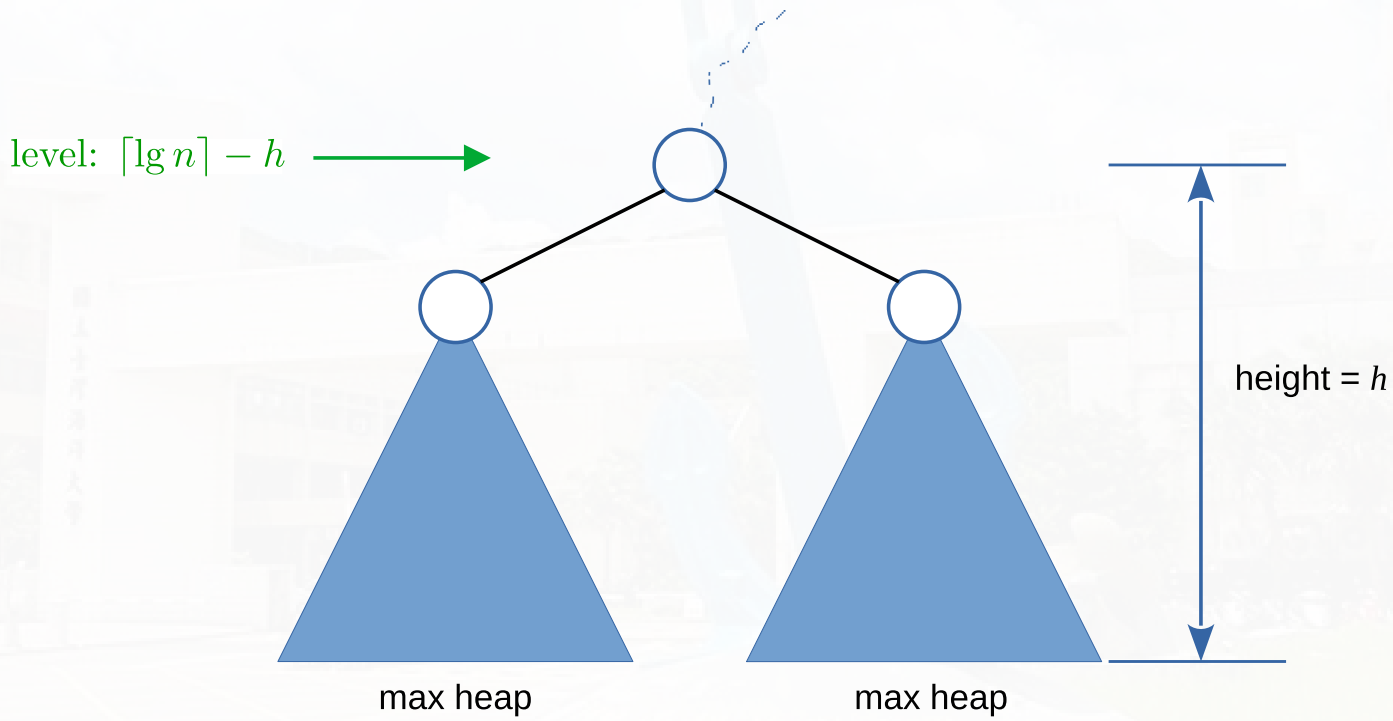


Heap recursive view (bottom-up)



- Assumption: the left subtree & the right subtree are max heaps.

Heap recursive view (bottom-up)



- Assumption: the left subtree & the right subtree are max heaps.
- Take the maximum of the roots of the two subtrees.

heapify(A, n, i) for a Max Heap (nonrecursive)

- Assumption: the left and right subtrees of i are already max heaps.

```
void heapify(element A[], int n, int i) {
    int parent = i;
    int child  = 2 * parent;  // default child: left child
    element temp = A[parent];
    // each loop goes down one level, so O(log n) time in a complete binary tree.
    while (child <= n) {
        /* choose the larger child */
        if (child < n && A[child].key < A[child + 1].key) child++;

        /* if temp already >= max(child), we're done */
        if (temp.key >= A[child].key) break;

        /* move the larger child up one level */
        A[parent] = A[child];
        parent = child;
        child  = 2 * parent;
    }
    A[parent] = temp;
}
```

heapify(A, n, i) for a Max Heap (recursive version)

```
void heapify(element A[], int n, int i) {
    int left = 2*i + 1, right = 2*i + 2; // a heap is a complete binary tree
    int largest = i;

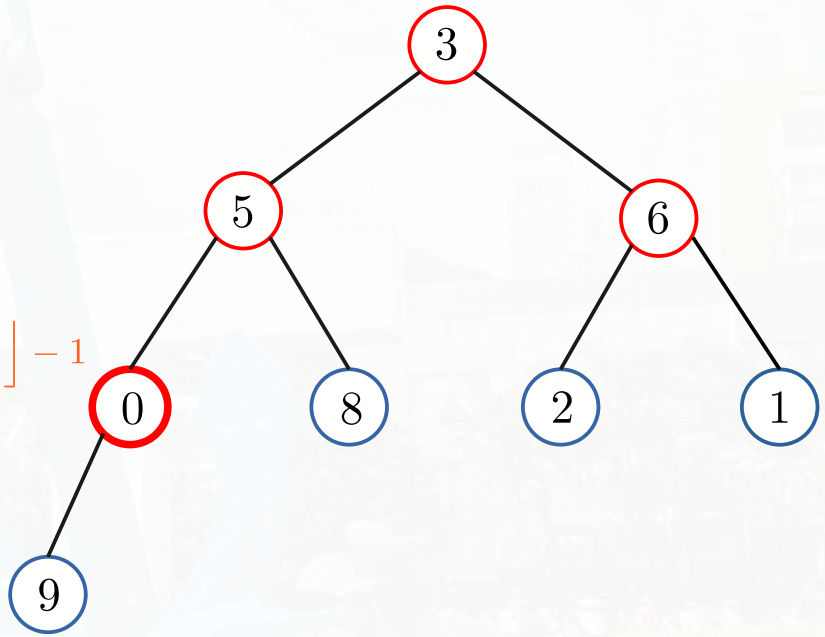
    if (left < n && A[left].key > A[largest].key)
        largest = left;
    if (right < n && A[right].key > A[largest].key)
        largest = right;

    if (largest != i) { // swap A[i] and A[largest]
        element tmp = A[i]; A[i] = A[largest]; A[largest] = tmp;
        heapify(A, n, largest);
    }
}
// IMPORTANT: We MUST heapify from the next level from the bottom
void buildMaxHeap(element A[], int n) {
    for (int i = n/2 - 1; i >= 0; --i)
        heapify(A, n, i);
}
```

Nodes to be Heapified

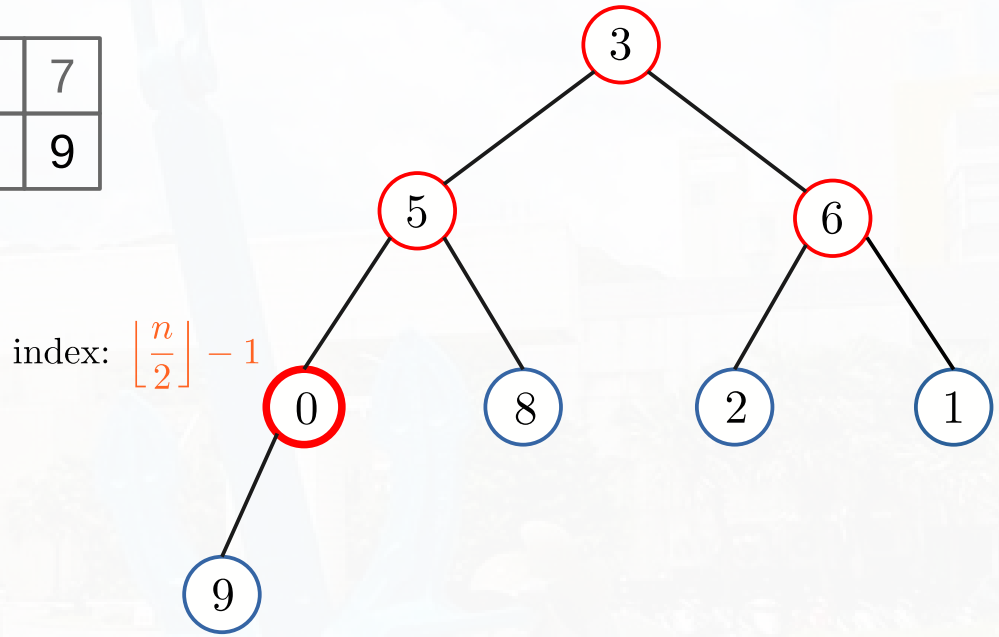
index	0	1	2	3	4	5	6	7
	3	5	6	0	8	2	1	9

index: $\left\lfloor \frac{n}{2} \right\rfloor - 1$



Nodes to be Heapified

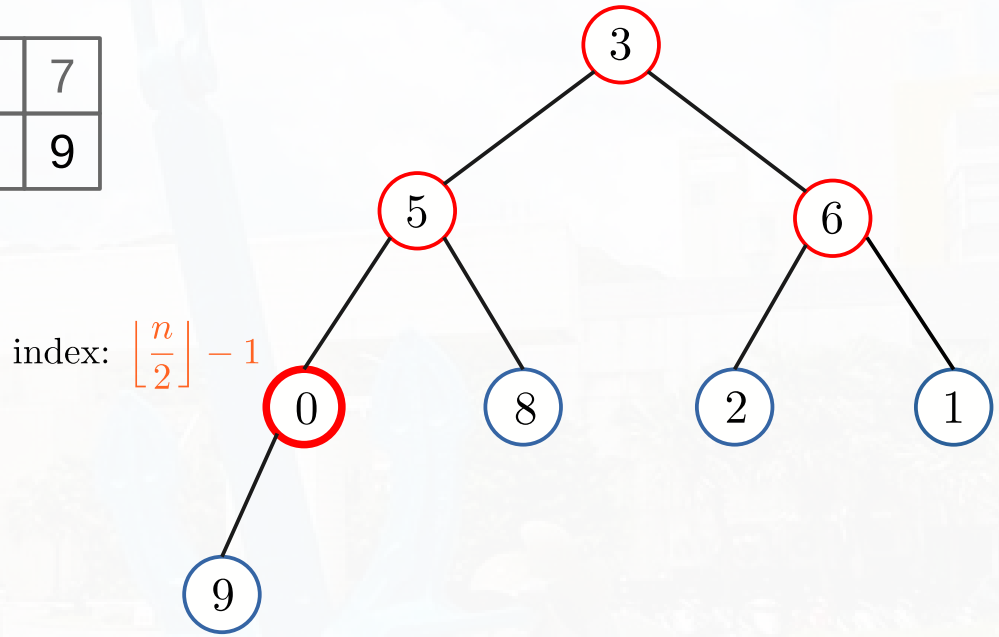
index	0	1	2	3	4	5	6	7
	3	5	6	0	8	2	1	9



- # Heapify steps: $\leq \sum_{h=1}^{\lg n - 1} h \cdot n_h =$

Nodes to be Heapified

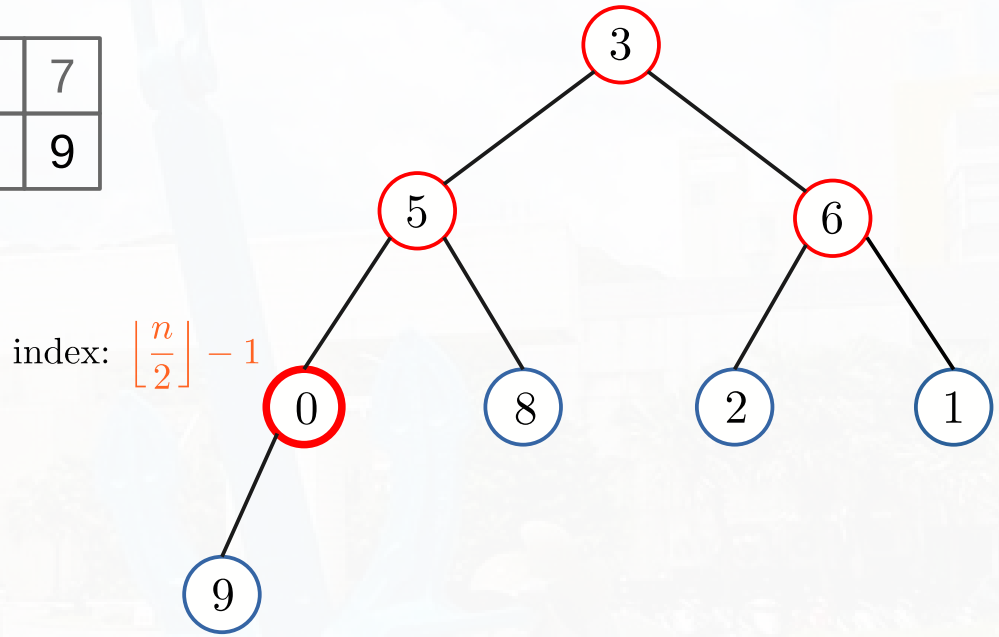
index	0	1	2	3	4	5	6	7
	3	5	6	0	8	2	1	9



- # Heapify steps: $\leq \sum_{h=1}^{\lg n - 1} h \cdot n_h = \sum_{h=1}^{\lg n - 1} h \cdot 2^{\lceil \lg n \rceil - 1 - h}$

Nodes to be Heapified

index	0	1	2	3	4	5	6	7
	3	5	6	0	8	2	1	9



- # Heapify steps: $\leq \sum_{h=1}^{\lg n-1} h \cdot n_h = \sum_{h=1}^{\lg n-1} h \cdot 2^{\lceil \lg n \rceil - 1 - h} \leq n \sum_{h=1}^{\lg n-1} \frac{h}{2^h}$.
  - n_h: the number of nodes at level h.

Exercise

Show that $\sum_{h=0}^{\infty} \frac{h}{2^h} = O(1)$.



Discussions